which we label A, B and O, where each of the three parts may be solved "exactly", with individual updates given as

$$\begin{aligned}\mathscr{U}_h^{\mathrm{A}}(\boldsymbol{q},\boldsymbol{p}) &= (\boldsymbol{q} + h\boldsymbol{M}^{-1}\boldsymbol{p}, \boldsymbol{p}),\\ \mathscr{U}_h^{\mathrm{B}}(\boldsymbol{q},\boldsymbol{p}) &= (\boldsymbol{q}, \boldsymbol{p} - h\nabla U(\boldsymbol{q})),\\ \mathscr{U}_h^{\mathrm{O}}(\boldsymbol{q},\boldsymbol{p}) &= (\boldsymbol{q}, e^{-\gamma h}\boldsymbol{p} + \sqrt{\mathrm{k_B T}\,(1 - e^{-2\gamma h})}\boldsymbol{M}^{1/2}\mathbf{R}),\end{aligned}$$

where $\mathbf{R}$ is a vector of i.i.d. normal random numbers. For the deterministic pieces A and B, the updates correspond to their flow maps, while the update for the O piece corresponds to the solution derived in (7.3). The relevant forward propagators for each piece of the Hamiltonian part are computed through the adjoint of the Lie derivative as

$$\mathcal{L}_{\mathrm{A}}^{\dagger} = -\boldsymbol{M}^{-1}\boldsymbol{p}\cdot\nabla_q, \qquad \mathcal{L}_{\mathrm{B}}^{\dagger} = \nabla U(\boldsymbol{q})\cdot\nabla_p.$$

Using this splitting strategy, we can define a family of schemes by the desired sequence of updates, codified as a string using the alphabet "ABO". For example, the method denoted $[\![\mathrm{ABO}]\!]$ (acting from left to right) has a resulting update

$$\mathscr{U}_h^{[\![\mathrm{ABO}]\!]} = \mathscr{U}_h^{\mathrm{O}} \circ \mathscr{U}_h^{\mathrm{B}} \circ \mathscr{U}_h^{\mathrm{A}}.$$

This corresponds to using the adjoint symplectic Euler scheme to solve the Newtonian part of the Langevin dynamics SDE, followed by an exact OU solve. An alternative is to use velocity Verlet for the Hamiltonian part, resulting in a scheme denoted $[\![\mathrm{BABO}]\!]$ with associated map

$$\mathscr{U}_h^{[\![\mathrm{BABO}]\!]} = \mathscr{U}_h^{\mathrm{O}} \circ \mathscr{U}_{h/2}^{\mathrm{B}} \circ \mathscr{U}_h^{\mathrm{A}} \circ \mathscr{U}_{h/2}^{\mathrm{B}}.$$

Note that the presence of two B characters in the string results in halving the corresponding piece's time step length. As we need the method to be consistent, we require that the each of the A, B and O pieces are propagated in time for a total time of h. As a result we use the convention that if a letter appears k times in a method's string, then each individual update piece uses a timestep h/k. For example, the scheme denoted $[\![\mathrm{OABOAOBAO}]\!]$ has an update

$$\mathscr{U}_h^{[\![\mathrm{OABOAOBAO}]\!]} = \mathscr{U}_{h/4}^{\mathrm{O}} \circ \mathscr{U}_{h/3}^{\mathrm{A}} \circ \mathscr{U}_{h/2}^{\mathrm{B}} \circ \mathscr{U}_{h/4}^{\mathrm{O}} \circ \mathscr{U}_{h/3}^{\mathrm{A}} \circ \mathscr{U}_{h/4}^{\mathrm{O}} \circ \mathscr{U}_{h/2}^{\mathrm{B}} \circ \mathscr{U}_{h/3}^{\mathrm{A}} \circ \mathscr{U}_{h/4}^{\mathrm{O}},$$

where the random numbers used in each O update are independent. This scheme will require a total of $4\mathrm{N}_c$ normal random numbers and two force evaluations per step (the force must be recomputed at the first B step following an A step; the number of force evaluations required is not directly related to the length of the string).

A class of methods like those considered above were discussed in [46] and given the general title of "Geometric" Langevin Algorithm (GLA). The idea of a GLA